

CMSE 801 - Day 6

Python Modules: Math and Plots

Sep 21, 2022

General Course Announcements

- Pre-class assignments for Monday are now due Saturday at 5:30 p.m. instead of Friday
- Pre-class assignments for Wednesday are now due Tuesday at 5:30 p.m.
- Homework 1 has been released and is due Wednesday Sep 28 at 11:59 p.m.

Questions/comments from Day 6 PCA

- For every output of plot, there is a line above the figure, what does it mean? I really don't like it because it makes me feel my code has some error, can I erase it?
- Is there an "automatic" way to make the legend not overlap the plots?
- Every time I import matplotlib, even when it was provided in Q1, when I run the code the kernel immediately dies.
- Perhaps just documenting how to get certain colors and line types
- Are we always going to use "in range(len" for getting the actual values in a list? I think I understand it better now, but it was a big question for me earlier in the course
- Do we need to write 'plt.show()'
- When I want to loop 'x' with the increment of '0.1' from 0 to 0.8, the output values for 'x' are [0, 0.1, 0.2, 0.30000000000000004, 0.4, 0.5, 0.6, 0.7, 0.7999999999999999]. Why this happens?
- When I modified the legend following the guidance, I got a warning like "Legend does not support", or my legend label becomes some error codes similar as the line above. If I keep it default, everything is okay.
- I think the in-class assignment are bit harder than the pre-class assignments. So I am taking a bit of time to write my answers, but my group is moving really quickly.
- Nope! Still enjoying it. It's hard but fun. / thanks :)

```
In [1]: # we import math module
import math

# define x, sin(x), cos(x)
xList = list()
sinList = list()
cosList = list()
x = 0
dx = 0.1
while x < 2*math.pi:
    xList.append(x)
```

```

sinList.append(math.sin(x))
cosList.append(math.cos(x))
x += dx

```

```

In [2]: print('x = ', xList)
        print('sin(x) = ', sinList)
        print('cos(x) = ', cosList)

```

```

x = [0, 0.1, 0.2, 0.30000000000000004, 0.4, 0.5, 0.6, 0.7, 0.7999999999999999,
0.8999999999999999, 0.9999999999999999, 1.0999999999999999, 1.2, 1.3, 1.40000000
00000001, 1.5000000000000002, 1.6000000000000003, 1.7000000000000004, 1.80000000
00000005, 1.9000000000000006, 2.0000000000000004, 2.1000000000000005, 2.20000000
00000006, 2.3000000000000007, 2.4000000000000001, 2.5000000000000001, 2.6000000000
000001, 2.7000000000000001, 2.8000000000000001, 2.9000000000000012, 3.00000000000000
013, 3.10000000000000014, 3.20000000000000015, 3.30000000000000016, 3.40000000000000
017, 3.50000000000000018, 3.6000000000000002, 3.7000000000000002, 3.800000000000000
2, 3.9000000000000002, 4.0000000000000002, 4.1000000000000001, 4.2000000000000001,
4.3000000000000001, 4.4, 4.5, 4.6, 4.699999999999999, 4.799999999999999, 4.899999
999999999, 4.999999999999998, 5.099999999999998, 5.1999999999999975, 5.299999999
999997, 5.399999999999997, 5.4999999999999964, 5.599999999999996, 5.69999999999
996, 5.799999999999995, 5.899999999999995, 5.999999999999995, 6.099999999999994,
6.199999999999994]
sin(x) = [0.0, 0.09983341664682815, 0.19866933079506122, 0.2955202066613396, 0.
3894183423086505, 0.479425538604203, 0.5646424733950354, 0.644217687237691, 0.71
73560908995227, 0.7833269096274833, 0.8414709848078964, 0.8912073600614353, 0.93
20390859672263, 0.963558185417193, 0.9854497299884603, 0.9974949866040544, 0.999
5736030415051, 0.9916648104524686, 0.973847630878195, 0.9463000876874144, 0.9092
974268256815, 0.8632093666488735, 0.8084964038195899, 0.7457052121767198, 0.6754
631805511504, 0.5984721441039558, 0.5155013718214634, 0.42737988023382895, 0.334
98815015590383, 0.23924932921398112, 0.14112000805986588, 0.04158066243328916, -
0.05837414342758141, -0.15774569414324996, -0.25554110202683294, -0.350783227689
6215, -0.44252044329485407, -0.5298361409084948, -0.6118578909427207, -0.6877661
591839753, -0.7568024953079294, -0.8182771110644114, -0.8715757724135886, -0.916
1659367494552, -0.9516020738895161, -0.977530117665097, -0.9936910036334644, -0.
9999232575641008, -0.9961646088358408, -0.9824526126243328, -0.958924274663139,
-0.9258146823277331, -0.8834546557201545, -0.8322674422239027, -0.77276448755598
94, -0.7055403255703945, -0.6312666378723243, -0.5506855425976414, -0.4646021794
137613, -0.37387666483024096, -0.27941549819893097, -0.18216250427210112, -0.083
0894028175026]
cos(x) = [1.0, 0.9950041652780257, 0.9800665778412416, 0.955336489125606, 0.921
0609940028851, 0.8775825618903728, 0.8253356149096783, 0.7648421872844885, 0.696
7067093471655, 0.6216099682706646, 0.5403023058681399, 0.4535961214255775, 0.362
3577544766736, 0.26749882862458735, 0.16996714290024081, 0.07073720166770268, -
0.029199522301289037, -0.12884449429552508, -0.22720209469308753, -0.32328956686
350396, -0.4161468365471428, -0.5048461045998579, -0.5885011172553462, -0.666276
0212798247, -0.737393715541246, -0.8011436155469343, -0.8568887533689478, -0.904
0721420170617, -0.9422223406686585, -0.9709581651495908, -0.9899924966004456, -
0.9991351502732795, -0.998294775794753, -0.9874797699088647, -0.966798192579460
5, -0.9364566872907957, -0.8967584163341462, -0.8481000317104072, -0.79096771191
44154, -0.7259323042001387, -0.6536436208636106, -0.5748239465332677, -0.4902608
213406987, -0.4007991720799746, -0.30733286997841935, -0.21079579943077972, -0.1
1215252693505487, -0.012388663462891449, 0.08749898343944551, 0.186512369422574,
0.2836621854632246, 0.3779777427129786, 0.4685166713003748, 0.5543743361791585,
0.6346928759426319, 0.7086697742912575, 0.7755658785102473, 0.8347127848391573,
0.8855195169413169, 0.9274784307440339, 0.9601702866503645, 0.9832684384425836,
0.9965420970232169]

```

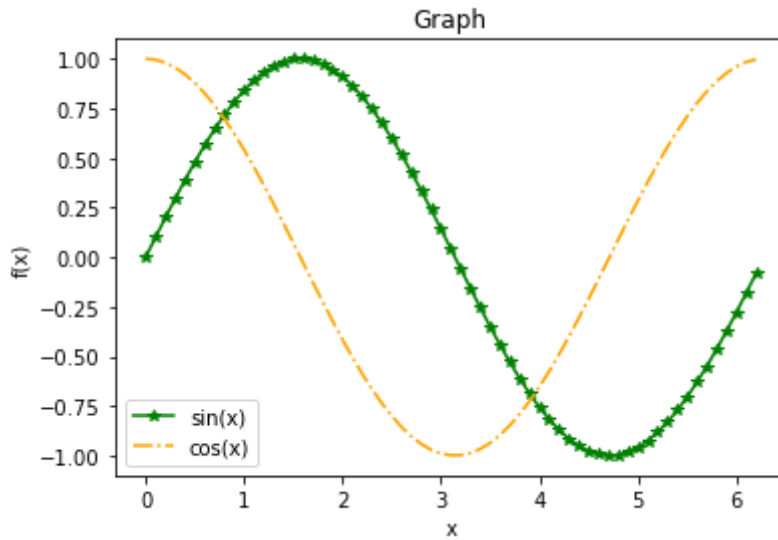
```

In [3]: import matplotlib.pyplot as plt
        %matplotlib inline

        plt.plot(xList,sinList,"g",label='sin(x)',marker="*",linestyle="-")
        plt.plot(xList,cosList,color="orange", label="cos(x)",linestyle="-.")
        plt.legend()
        plt.xlabel("x")

```

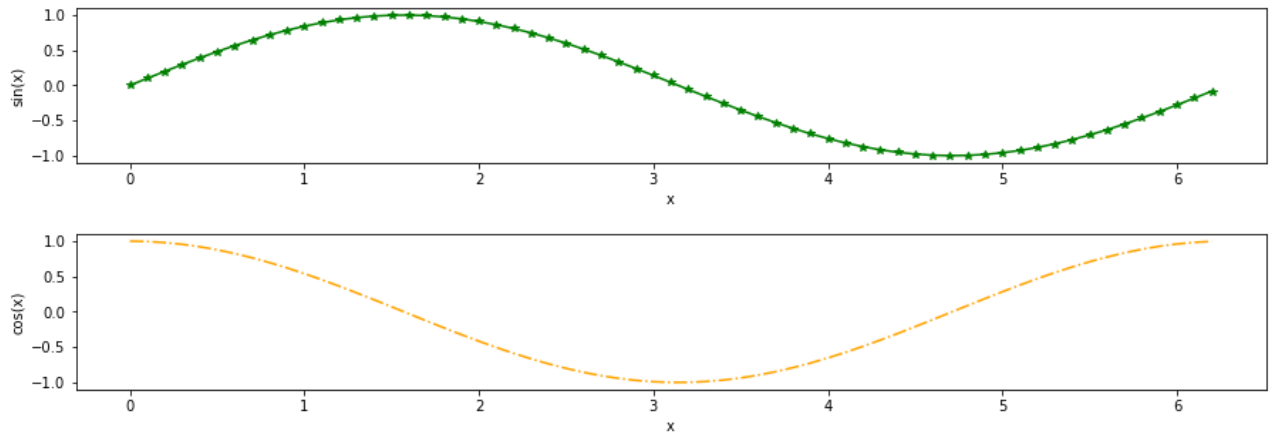
```
plt.ylabel("f(x)")
plt.title("Graph")
plt.show()
```



```
In [4]: plt.figure(figsize=[15,2])
plt.plot(xList,sinList,"g",marker="*",linestyle="-")
plt.xlabel("x")
plt.ylabel("sin(x)")

plt.figure(figsize=[15,2])
plt.plot(xList,cosList,color="orange",linestyle="-.")
plt.xlabel("x")
plt.ylabel("cos(x)")
```

Out[4]: Text(0, 0.5, 'cos(x)')



```
In [5]: figure, axes = plt.subplots(2,1,figsize=[12,3])
axes[0].plot(xList,sinList,color="g")
axes[0].set_xlabel('x')
axes[0].set_ylabel('sin(x)')
axes[1].plot(xList,cosList,color="orange")
axes[1].set_xlabel('x')
axes[1].set_ylabel('cos(x)')
```

Out[5]: Text(0, 0.5, 'cos(x)')

